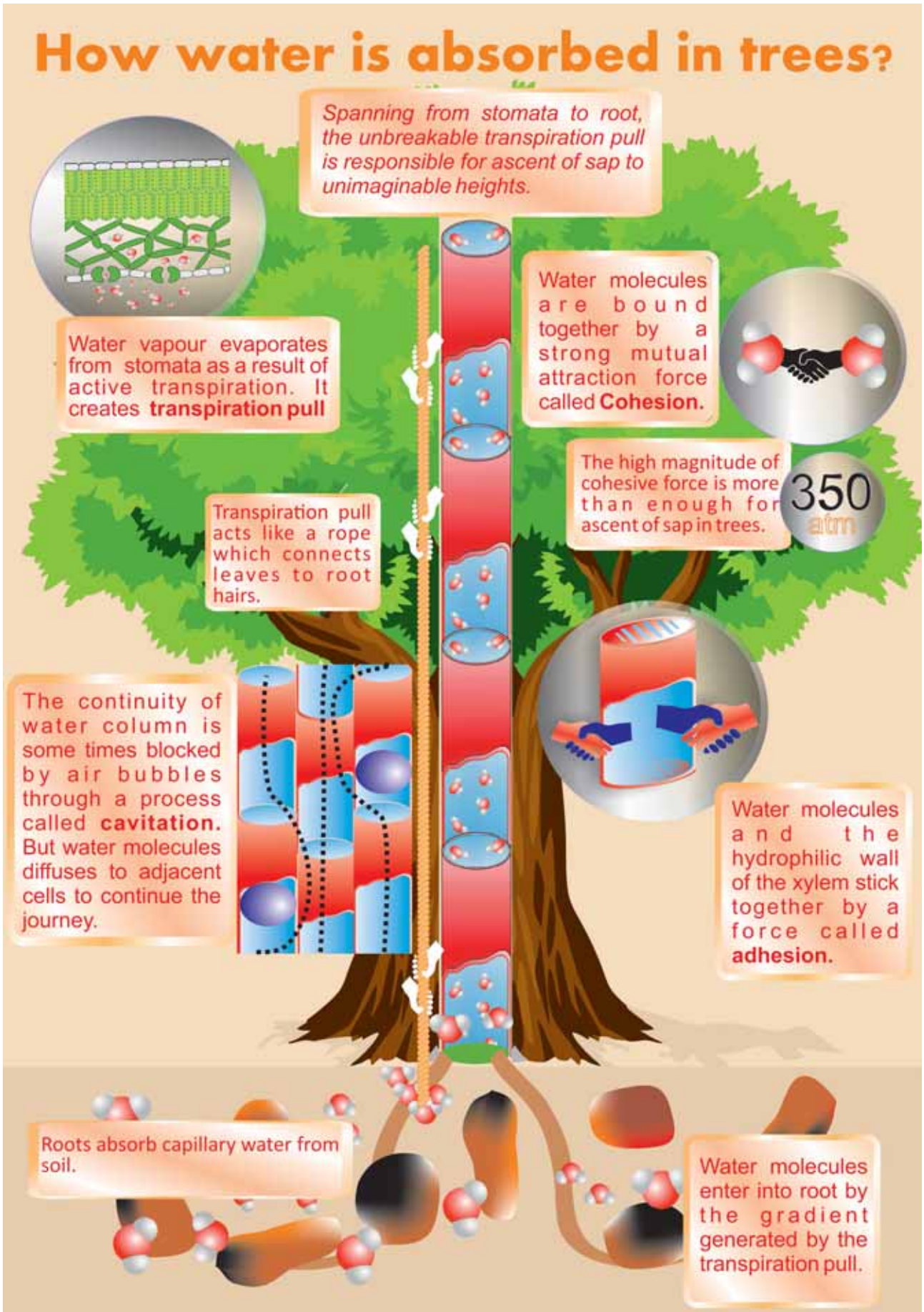

How water is absorbed in trees?
Spanning from stomata to root, the unbreakable transpiration pull is responsible for ascent of sap to unimaginable heights.
Water vapour evaporates from stomata as a result of active transpiration. It creates transpiration pull
Water molecules are bound together by a strong mutual attraction force called Cohesion.
The high magnitude of cohesive force is more than enough for ascent of sap in trees.
350 atm
Transpiration pull acts like a rope which connects leaves to root hairs.
The continuity of water column is some times blocked by air bubbles through a process called cavitation. But water molecules diffuses to adjacent cells to continue the journey.
Water molecules and the hydrophilic wall of the xylem stick together by a force called adhesion.
Roots absorb capillary water from soil.
Water molecules enter into root by the gradient generated by the transpiration pull.